

# Project Synopsis

## SmartExp – Chat-Assisted Expense Tracker Using NLP

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### 1. Title of the Project

SmartExp – Chat-Assisted Expense Tracker Using Natural Language Processing

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### 2. Introduction

Managing personal expenses is an essential but often neglected task in daily life. Many individuals fail to track their spending consistently due to the repetitive and manual nature of traditional expense tracking applications. As a result, users lack clear visibility into their financial habits and spending patterns.

SmartExp is a web-based expense tracking system that integrates a conversational chatbot interface with a visual analytics dashboard. The system allows users to manage expenses not only through standard forms but also through natural language commands, thereby improving usability and reducing manual effort.

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### 3. Problem Statement

Most existing expense tracking applications rely heavily on manual data entry and rigid interfaces. These limitations reduce user engagement and make consistent expense tracking difficult. Additionally, many systems fail to provide intuitive interaction methods or meaningful insights into spending behavior.

There is a need for a system that:

- Simplifies expense recording
  - Reduces user effort through conversational interaction
  - Provides clear visual insights
  - Maintains data security and usability
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### 4. Objectives of the Project

The main objectives of the project are:

- To design and develop a secure expense tracking system
- To implement a chatbot interface using NLP for natural language interaction
- To classify user intents using machine learning techniques

- To extract relevant entities such as amount, category, and date from user input
  - To visualize expense data using interactive charts
  - To demonstrate the practical application of NLP and ML concepts
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## 5. Scope of the Project

The scope of SmartExp includes:

- User authentication and profile management
- Expense and budget management
- Chat-based interaction for expense operations
- Dashboard-based analytics and summaries
- Responsive design for mobile and desktop devices

The project is limited to commonly used expense categories and predefined date expressions.

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## 6. Methodology

The system follows a **hybrid NLP approach**:

- **Machine Learning** is used for intent classification using TF-IDF vectorization and Logistic Regression.
- **Rule-based techniques** are used for entity extraction such as amount, category, and date.

The application follows a client-server architecture:

- Frontend developed using React.js
  - Backend developed using FastAPI
  - Database used for persistent storage
  - APIs used for communication between frontend and backend
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## 7. Tools and Technologies Used

### Frontend:

React.js, Axios, Chart.js, CSS Modules

### Backend:

FastAPI, SQLAlchemy, JWT Authentication

### Database:

PostgreSQL / SQLite

### NLP & Machine Learning:

Scikit-learn, NLTK, Regular Expressions

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## 8. Applications

- Personal expense management
- Academic demonstration of NLP and ML integration
- Portfolio project for full-stack development

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## 9. Limitations

- Expense categorization is keyword-based
- Date parsing supports limited expressions
- No predictive analysis or forecasting
- No OCR or voice-based input

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## 10. Future Enhancements

- Advanced NLP using embedding-based models
- Expense prediction using time-series analysis
- Receipt scanning using OCR
- Voice-enabled interaction
- Anomaly detection for unusual spending

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## 11. Conclusion

SmartExp demonstrates how a modern expense tracking system can be enhanced using natural language interaction and basic machine learning techniques. The project successfully integrates frontend development, backend services, and NLP concepts into a cohesive and user-friendly application. It serves as a strong academic project with scope for further enhancement.

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## 12. Student Details

**Project Title:** SmartExp

**Group Number:**7

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**Course:** Bsc computer science

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